INVERSE TRIGONOMETRIC FUNCTION

SUBJECT: MATHEMATICS **DATE: 18-03-2024**

1. Which of the following options is/are TRUE?

(A) $4\tan^{-1}\frac{1}{5}-\tan^{-1}\frac{1}{239}=\frac{\pi}{4}$ (B) $\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{8}=\frac{\pi}{4}$

(C) $\cos\left(2\tan^{-1}\frac{1}{7}\right)=\sin\left(-4\tan^{-1}\frac{1}{3}\right)$ (D) $\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{7}{25}=\cos^{-1}\frac{253}{325}$

2. The solution set of $\tan^{-1}\frac{x-1}{x-2}+\tan^{-1}\frac{x+1}{x+2}=\frac{\pi}{4}$ is

(A) $[-\sqrt{2},\sqrt{2}]$ (B) $(-\sqrt{2},\sqrt{2})$ (C) $\{-\sqrt{2},\sqrt{2}\}$ (D) None of these

3. The product of all values of x satisfying the equation $\sin^{-1}\cos\left(\frac{2x^2+10|x|+4}{x^2+5|x|+3}\right)=\cos\left(\cot^{-1}\left(\frac{2-18|x|}{9|x|}\right)\right)+\frac{\pi}{2}$ is

(A) 9 (B) –9 (C) –3 (D) –1

4. Let $\begin{vmatrix}\tan^{-1}x & \tan^{-1}2x & \tan^{-1}3x\\ \tan^{-1}3x & \tan^{-1}x & \tan^{-1}2x\\ \tan^{-1}2x & \tan^{-1}3x & \tan^{-1}x\end{vmatrix}=0$, then the number of values of x satisfying the equation is

(A) 1 (B) 2 (C) 3 (D) 4

5. $\sin^{-1}(\sin 5)>x^2-4x$ holds if

(A) $x=2-\sqrt{9-2\pi}$ (B) $x=2+\sqrt{9-2\pi}$

(C) $x>2+\sqrt{9-2\pi}$ (D) $x\in(2-\sqrt{9-2\pi},2+\sqrt{9-2\pi})$

6. $\sum_{r=1}^{n}\sin^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{\sqrt{r(r+1)}}\right)$ is equal to

(A) $\tan^{-1}(\sqrt{n})-\frac{\pi}{4}$ (B) $\tan^{-1}(\sqrt{n+1})-\frac{\pi}{4}$

(C) $\tan^{-1}(\sqrt{n})$ (D) $\tan^{-1}(\sqrt{n+1})$

7. If $\sin^{-1}a+\sin^{-1}b+\sin^{-1}c=\pi$, then the value of $a\sqrt{(1-a^2)}+b\sqrt{(1-b^2)}+c\sqrt{(1-c^2)}$ will be

(A) 2 abc (B) abc (C) $\frac{1}{2}abc$ (D) $\frac{1}{3}abc$

8. The value of $\tan\left(\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)\tan\left(\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)\right)$, where $x\in(0,1)$, is equal to

(A) 0 (B) 1 (C) –1 (D) None of these

9. The maximum value of $f(x)=\tan^{-1}\left(\frac{(\sqrt{12}-2)x^2}{x^4+2x^2+3}\right)$ is

(A) 18° (B) 36° (C) 22.5° (D) 15°

10. The least and the greatest values of $\left(\sin^{-1}x\right)^3+\left(\cos^{-1}x\right)^3$ are

(A) $\frac{-\pi}{2},\frac{\pi}{2}$ (B) $\frac{-\pi^3}{8},\frac{\pi^3}{8}$ (C) $\frac{\pi^3}{32},\frac{7\pi^3}{8}$ (D) None of these

11. If $\sin^{-1}\left(a-\frac{a^2}{3}+\frac{a^3}{9}+\cdots\right)+\cos^{-1}\left(1+b+b^2+\cdots\right)=\frac{\pi}{2}$, then

(A) $b=\frac{2a-3}{3a}$ (B) $b=\frac{3a-2}{2a}$ (C) $a=\frac{3}{2-3b}$ (D) $a=\frac{2}{3-2b}$

12. If $\left(\sin^{-1}x+\sin^{-1}w\right)\left(\sin^{-1}y+\sin^{-1}z\right)=\pi^2$, then $D=\begin{vmatrix} x^{N_1} & y^{N_2} \\ z^{N_3} & w^{N_4}\end{vmatrix}\left(N_1,N_2,N_3,N_4\in N\right)$

(A) Has a maximum value of 2 (B) Has a minimum value of 0
(C) 16 different D are possible (D) Has a minimum value of –2

13. Area bounded by $\cos^{-1}(\cos x)$ and $\sin x, x\in[0,2\pi]$ is

(A) π^2-2 (B) π^2 (C) π^2+2 (D) π^2+4

14. Complete set of values of real k such that equation $\tan^{-1}x-\cos^{-1}x=k$ has a real solution is

(A) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ (B) $\left[-\frac{5\pi}{2},\frac{\pi}{2}\right]$ (C) $\left(-\frac{\pi}{4},\frac{\pi}{4}\right)$ (D) $\left[-\frac{5\pi}{4},\frac{\pi}{4}\right]$

15. If $f:[-1,1]\to\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ is defined as $f(x)=\sin^{-1}x$ and $fo\,fo\,f(x)=g(x)$, then

(A) domain of $g(x)$ is $[-\sin(\sin 1),\sin(\sin 1)]$ (B) $g^{-1}(x)=\sin x$

(C) $g(x)$ is monotonic (D) $g(x)$ has a range $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$